

Solutions to First Problem Sheet

on
Complex Analysis

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Topics: Complex Arithmetic
Argand Diagram

“The shortest path between two truths in the real domain passes through the complex domain.”

-Jacques Hadamard

Problem 1

Recall that $\mathbb{R} = \mathbf{C}/\sim$, where $\mathbf{C}$ is the collection of all Cauchy sequences in $\mathbb{Q}$ and $\sim$ is the equivalence relation defined via

$$(x_n) \sim (y_n) \iff [\forall \varepsilon \in \mathbb{Q}^+ \exists K \in \mathbb{N} : n \geq K \implies |x_n - y_n| < \varepsilon].$$

For each $q \in \mathbb{Q}$, let

$$\hat{q} \equiv (q, q, \dots)$$

denote the constant sequence with value q .

- i) Show that $\hat{q} \in \mathbf{C}$ for every $q \in \mathbb{Q}$, i.e. every constant sequence is Cauchy.
- ii) Show that $\hat{q}$ converges to q in $\mathbb{Q}$ in the sense of the Euclidean metric.
- iii) Using ii), prove that $\hat{p} \sim \hat{q}$ if and only if $p = q$, and thus propose a canonical embedding of $\mathbb{Q}$ into $\mathbb{R}$, i.e., a map $f : \mathbb{Q} \hookrightarrow \mathbb{R}$ such that $f(\mathbb{Q}) \subseteq \mathbb{R}$.

We usually write $\mathbb{Q} \subseteq \mathbb{R}$ to denote the fact that $f(\mathbb{Q}) \subseteq \mathbb{R}$.

Solution 1

- i) Given $\varepsilon \in \mathbb{Q}^+$ let $\hat{q}_l$ be the l -th element of the sequence defined above (in this case, $\hat{q}_l = q$ for all l) and let $K = 1$. Then for $n, m \geq K$, the elements of the sequence $\hat{q}$ are such that

$$|\hat{q}_n - \hat{q}_m| = |q - q| = 0 < \varepsilon$$

for any $\varepsilon \in \mathbb{Q}^+$ so that the sequence is Cauchy.

- ii) Given $\varepsilon \in \mathbb{Q}^+$, choose $K = 1$ so that

$$d(\hat{q}_n, q) = |\hat{q}_n - q| = |q - q| = 0 < \varepsilon$$

and thus $\hat{q} \rightarrow q$.

- iii) ($\Rightarrow$) If $\hat{p} \sim \hat{q}$, then given any $\varepsilon \in \mathbb{Q}^+$ there is some $K \in \mathbb{N}$ such that $|\hat{p}_n - \hat{q}_n| < \varepsilon$ for $n \geq K$, but this just means that $|p - q| < \varepsilon$ for all $\varepsilon \in \mathbb{Q}^+$. Next, suppose that $p \neq q$, then $p - q \neq 0$, then $|p - q| > 0$ and thus by the Archimedean property we get that there is some rational number r such that $0 < r < |p - q|$, but then for $\varepsilon = r$ the assumption implies that $r < |p - q| < r$, clearly, a contradiction. We therefore conclude that $p = q$. Conversely, if $p = q$, it is readily seen that the sequences are related through $\sim$. The canonical embedding is clearly $f(q) = [\hat{q}] \equiv [(q, q, \dots)]$, which sends each rational number to the constant (Cauchy, by the first bullet) sequence defined by it.



Problem 2

The definition of the order relation denoted by $>$ in the real number system is based upon the existence of a subset $\mathbb{P}$ having the following properties:

- i) For any number $a \neq 0$, either a or $-a$ (but not both) belongs to $\mathbb{P}$.
- ii) If a and b belong to $\mathbb{P}$, then so does $a + b$.
- iii) If a and b belong to $\mathbb{P}$, then so does ab .

When such a set $\mathbb{P}$ exists we write $a > b$ if and only if $a - b$ belongs to $\mathbb{P}$. Prove that the complex numbers system does not possess a nonempty subset $\mathbb{P}$ having properties i), ii), iii).

Solution 2

Given that $i \neq 0$, we may assume either $i \in \mathbb{P}$ or $-i \in \mathbb{P}$. Both assumptions lead to a contradiction in i) using iii). $\star$

Problem 3

Show that if $(z^*)^2 = z^2$, then z is either real or pure imaginary (is the converse also true?). Prove that $(z^*)^k = (z^k)^*$ for every integer k (provided $z \neq 0$ when $k \leq 0$).

Solution 3

Let $z = x + iy$, then $(z^*)^2 = z^2$ if and only if $xy = 0$ which is true if and only if $x = 0$ or $y = 0$. To prove the next proposition proceed by induction, it is clearly true for $k = 1$, then if it is true for some m , we have

$$(z^*)^m = (z^m)^* \implies (z^*)^{m+1} = (z^m)^* z^* = (z^m z)^* = (z^{m+1})^*.$$

$\star$

Problem 4

Show that $\Re(z + w) = \Re(z) + \Re(w)$. Give an example to show that $\Re(zw) \neq \Re(z)\Re(w)$ and thus find the correct equation for $\Re(zw)$. Using your formula deduce that $\Re(zw) = \Re(z)\Re(w)$ if and only if either z or w is real.

Solution 4

The first part directly follows from the definition of sum of complex numbers, for the counterexample we take $z = i, w = -i$ so that $\Re(z) = \Re(w) = 0$ and $\Re(zw) = 1$. The general formula is

$$\Re(zw) = \Re(z)\Re(w) - \Im(z)\Im(w)$$

so that the previous equality holds if and only if $\Im(z)\Im(w) = 0$ so either z is real or w is. $\star$

Problem 5

Write the following numbers in the Cartesian form $a + ib$ with $a, b \in \mathbb{R}$.

- i) $4i(2 - i)^2$;
- ii) $(1/2 + \sqrt{3}i/2)^3$;
- iii) $(i - \pi)/(i + \pi)$;
- iv) $(x + iy)/(x - iy)$ for $x, y \in \mathbb{R}$;
- v) $(2 - i)(3 + i)(4 + i)^2/(1 + i)^*$

Solution 5

- i) $-16 - 12i$
- ii) -1
- iii) $(1 - \pi^2)/(1 + \pi^2) + i2\pi/(1 + \pi^2)$
- iv) $(x^2 - y^2)/(x^2 + y^2) + i2xy/(x^2 + y^2)$
- v) $36 + 77i$



Problem 6

Find two numbers whose sum is 10 and whose product is 40. (This problem is of some historical value. It is said that Girolamo Cardan (1501-1576) first stumbled upon complex numbers while solving it.)

Solution 6

We seek to find numbers such that $z + w = 10$ and $zw = 40$ so multiplying the first equation by z we get that $z^2 + zw = 10z$ which by the second equation, becomes $z^2 - 10z + 40 = 0$. There is no real number satisfying such equation as can be seen from factorising it as

$$(z - 5)^2 = -65,$$

but with the complex numbers at our disposal, we can propose the solutions as

$$z = 5 + i\sqrt{65}, \quad z' = 5 - i\sqrt{65},$$

which are clearly complex.



Problem 7

Find a general formula for i^n with $n \in \mathbb{Z}$.

Solution 7

The formula is readily verified to be

$$\begin{cases} i^{2k} = (-1)^k & \text{for } n = 2k, \\ i^{2k+1} = (-1)^k i & \text{for } n = 2k + 1. \end{cases}$$



Problem 8

Deduce the appropriate identities for complex conjugation concerning the sum, difference, multiplication and division of complex numbers and hence show that:

- a) $|z_1 z_2| = |z_1| |z_2|$.
- b) $|z_1/z_2| = |z_1|/|z_2|$ provided $z_2 \neq 0$.

Solution 8

- a) $|z_1 z_2|^2 = z_1 z_2 z_1^* z_2^* = z_1 z_1^* z_2 z_2^* = |z_1|^2 |z_2|^2$.
- b) $\left| \frac{z_1}{z_2} \right|^2 = \frac{z_1 z_1^*}{z_2 z_2^*} = \frac{|z_1|^2}{|z_2|^2}$ provided that $z_2 \neq 0$.



Problem 9

Show that $|\Re(z)| \leq |z|$, $|\Im(z)| \leq |z|$ and $|z + w|^2 = |z|^2 + |w|^2 + 2\Re(zw^*)$. Hence show that

$$|z + w| \leq |z| + |w|.$$

This is known as the **triangle inequality**.

Solution 9

Let $z = x + iy$, then

$$|z| = \sqrt{x^2 + y^2} \geq \sqrt{x^2} = |x| \quad [\text{respectively, } \sqrt{y^2} = |y|],$$

so that $|z| \geq |\Re(z)|$ [respectively, $|z| \geq |\Im(z)|$].

For the next part, compute

$$|z + w|^2 = |z|^2 + |w|^2 + z^*w + w^*z,$$

and since $q + q^* = 2\Re(q)$ for any $q \in \mathbb{C}$, we get the desired result. Finally, combining the former and the latter, we see that

$$\begin{aligned} |z + w|^2 &= |z|^2 + |w|^2 + 2\Re(zw^*) \leq |z|^2 + |w|^2 + 2|\Re(zw^*)| \leq |z|^2 + |w|^2 + 2|z||w^*| \\ &= |z|^2 + |w|^2 + 2|z||w| = (|z| + |w|)^2, \end{aligned}$$

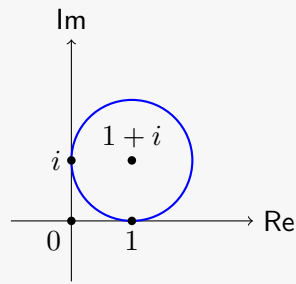
and the triangle inequality now readily follows. ☆

Problem 10

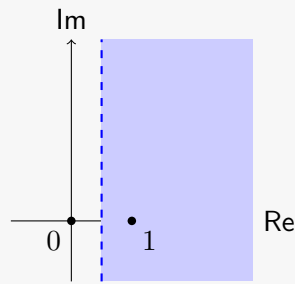
Identify and sketch the set points satisfying

- a) $|z - 1 - i| = 1$.
- b) $|z - 1| < |z|$.
- c) $\Re(iz + 2) > 0$.
- d) $|z - z_0| = |z - z_1|$, $z_0 = -1 + i$, $z_1 = 1 + 2i$.

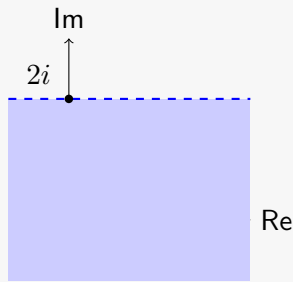
Solution 10



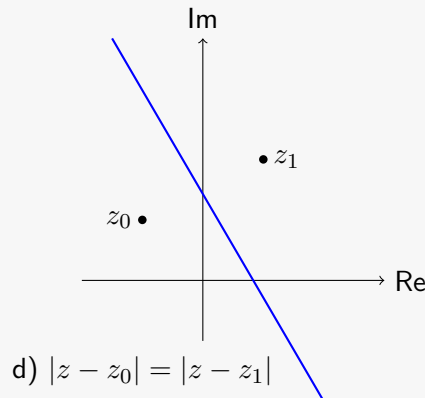
a) $|z - 1 - i| = 1$



b) $|z - 1| < |z|$



c) $\Re(iz + 2) > 0$



d) $|z - z_0| = |z - z_1|$



Problem 11

What is an upper bound for $|z - 3|$ if $|z - i| \leq 1$?

Solution 11

$|z - 3| \leq |z| + 3$ and since $|z| - |i| \leq 1$, then $|z| \leq 2$ so a bound for $|z - 3|$ is 5.



Problem 12

Find an upper bound for $|z - i|$ such that $|3z - 3i| < \varepsilon$ for $\varepsilon > 0$ given. What would be a bound such that $|z^2 + 1| < \varepsilon$?

Solution 12

The first bound would be $\varepsilon/3$ and the second one would be $\inf\{\varepsilon/3, 1\}$.



Problem 13

Let n be a natural number and let $\{a_k\}_{k=1}^n$ be complex numbers with $a_n \neq 0$. The expression

$$p(z) = a_0 + a_1z + a_2z^2 + \cdots + a_nz^n$$

is called a **polynomial** in z . Show that for any polynomial p of degree $n \geq 1$, one can write

$$p(z) = p(z_0) + h(z)(z - z_0)$$

where h is another polynomial of degree $n - 1$.

Solution 13

Given $p(z) = a_k z^k$, we take $p(z) - p(z_0)$

$$a_k z^k - a_k z_0^k = \sum_{k=0}^n a_k (z^k - z_0^k) = \sum_{k=0}^n a_k (z - z_0) \left(\sum_{j=1}^k z^{k-j} z_0^{j-1} \right) = (z - z_0) h(z),$$

where

$$h(z) = \sum_{k=0}^n \sum_{j=1}^k a_k \left(z^{k-j} z_0^{j-1} \right)$$

is a polynomial of degree $n - 1$.

